

Answer all questions

1. A simplified energy level system is given for a particular four-level laser system. Electrons are pumped (by means of infra-red radiation) from the ground state to level P, and drop to U, setting up a population inversion.

level P _____
level U _____ 0.820 eV

level L _____ 0.051 eV
ground state _____ 0

(a) (i) Calculate the wavelength of radiation emitted in the transition from level U to level L. [3]

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(ii) Explain how stimulated emission enables amplification of infra-red radiation of this wavelength. [3]

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(b) Explain the advantage of a four-level laser system over a three-level system. [2]

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